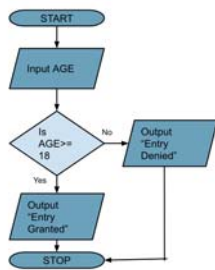


B 1.1.4 Flowcharts



B 1.1.4 Trace flowcharts for a range of programming algorithms.

A flowchart is a visual representation of an algorithm using standardised symbols connected by arrows. Tracing a flowchart means following its execution step by step, recording how variable values change at each stage and determining what output the algorithm produces for a given input. This is a core examination skill.

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Key Questions

- What are the six standard flowchart symbols and what does each one represent?
- How do you trace a flowchart to determine the output of an algorithm for a given input?
- How is a trace table structured, and what information does each row record?
- What is the purpose of a Connector symbol and when is it used?

IB Command Term

Trace

Follow and record the effects of an algorithm step by step.

In practice this means working through a flowchart symbol by symbol, updating variable values in a trace table as you go, and recording the final output.

What to Include

A complete trace table showing the value of every variable after each step that changes it, the result of every Decision, and the final output. Empty cells indicate that a variable was not changed at that step.

Standard flowchart symbols

The IB Computer Science syllabus specifies six standard symbols. These must be used correctly in any flowchart you draw or interpret.

SYMBOL NAME	SHAPE	PURPOSE AND RULES
Start/End		Marks where the algorithm begins or terminates. Every flowchart has exactly one Start and one End.
Process / Operation		Represents any action performed by the algorithm: a calculation, an assignment of a value to a variable, or any other operation.
Decision		Represents a condition that evaluates to true or false (yes or no). The flow branches into two paths depending on the result.
Input / Output		Represents data entering the system (user input) or data leaving the system (displayed output or printed results).

SYMBOL NAME	SHAPE	PURPOSE AND RULES
Flowline		Indicates the direction of flow between symbols. Every symbol is connected by at least one flowline; arrows must be labelled Yes/No on Decision branches.
Connector		Used to connect sections of a flowchart that are on separate pages, or to avoid crossing flow lines. Contains a matching letter or number at each end of the connection.

COMMON ERRORS TO AVOID

Using an oval instead of a rounded rectangle for Start/End - the IB syllabus specifies a rounded rectangle (also called a stadium or pill shape).

Labelling Decision branches without Yes/No (or True/False) - both exits from a diamond must be labelled.

Confusing Input/Output (parallelogram) with Process (rectangle) - reading a value from the user is an input operation, not a process.

Omitting the Connector when a flowchart continues on a second page - unpaired connectors make the flow ambiguous.

How to trace a flowchart

Tracing is a systematic process. Follow these steps every time.

1	Identify all variables used in the flowchart. Create a trace table with one column per variable, one column per Decision condition, and one column for Output. Add a row for each step that changes at least one value.
2	Begin at the Start symbol. Note the input value(s) given in the question and record them in the first row of the trace table.
3	Follow the flowline to the next symbol. Execute what that symbol instructs: update a variable value in the table (Process), record a new input (Input), or evaluate a condition (Decision).

4	At every Decision diamond, evaluate the condition using the current variable values. Record Yes or No in the Decision column. Follow the correct flowline based on the result.
5	Continue step by step until you reach the End symbol. Record any output produced by Output symbols in the Output column.
6	Check your final answer by reading the Output column. If the question asks what a specific variable equals at a certain point, read that column at that row.

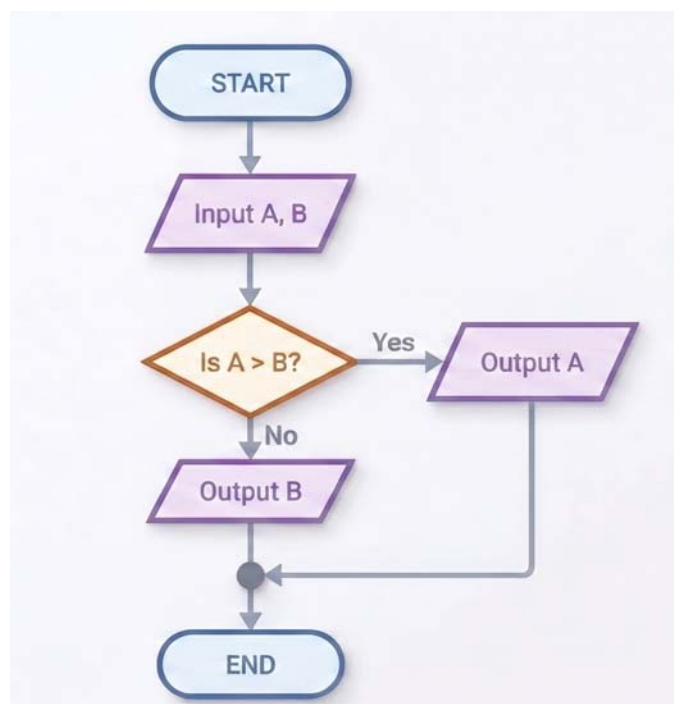
TRACE TABLE CONVENTIONS

Leave a cell empty if that variable was not changed in that step - do not copy the previous value forward. Write the value of a variable after the step that changes it, not before. If the algorithm contains a loop, each pass through the loop produces a new row in the trace table. Always include the row that records the final output, even if no variable values change in that row.

Worked example 1: finding the larger of two numbers

This example uses a simple flowchart with one Decision to compare two input values and output the larger one. It introduces the trace method without the added complexity of a loop.

THE ALGORITHM



Note: this flowchart has no loop, so the trace table has only a small number of rows. The Decision column records whether the condition $A > B$ evaluated to Yes or No.

Trace for input $A = 7, B = 3$

Step	A	B	$A > B?$	Output
Input A, B	7	3		
Decision: $A > B?$			Yes	
Output A				7

Because $A > B$ evaluates to Yes, the flow follows the Yes branch and outputs A (which is 7). The No branch is not taken and Output B is never reached.

Trace for input $A = 2, B = 9$

Step	A	B	$A > B?$	Output
Input A, B	2	9		
Decision: $A > B?$			No	
Output B				9

Because $A > B$ evaluates to No, the flow follows the No branch and outputs B (which is 9).

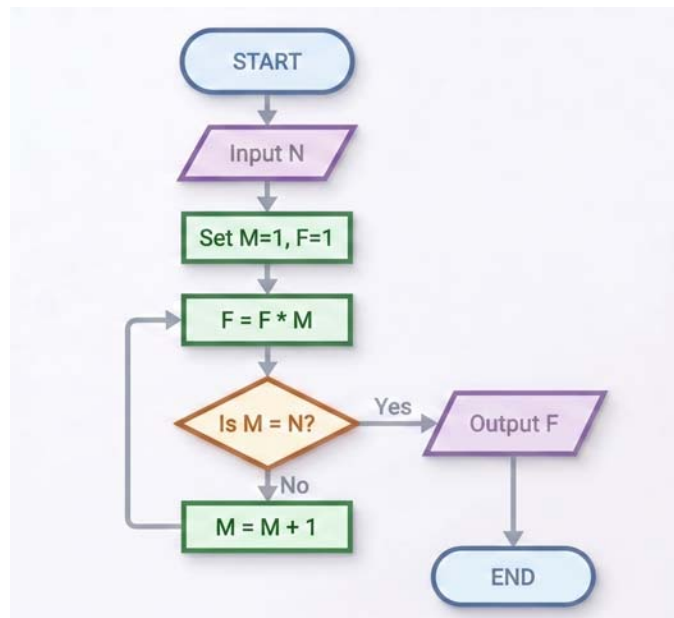
WHAT THIS EXAMPLE SHOWS

A trace table records only what changes at each step. The values of A and B do not change after input, so their cells are empty in the Decision and Output rows. Reading across any row gives you the current state of all variables at that moment in the execution.

Worked example 2: calculating a factorial

This example uses a flowchart with a loop (iteration). Loops produce multiple rows in the trace table -- one row per pass through the loop body.

THE ALGORITHM



The variable M is a counter that starts at 1 and increments by 1 each pass. F accumulates the factorial product. The loop continues until M equals N.

Trace for input N = 5

N	M	F	M = N?	Output
5	1	1		
		1	No	
	2	2	No	
	3	6	No	
	4	24	No	
	5	120	Yes	120

Reading through the trace row by row:

- Row 1: N is input as 5. M and F are both initialised to 1.
- Row 2: $F = F * M = 1 * 1 = 1$. $M = N?$ is $1 = 5$, which is No. Loop continues.
- Row 3: M increments to 2. $F = F * M = 1 * 2 = 2$. $M = N?$ is $2 = 5$, No.

- Row 4: M increments to 3. $F = F * M = 2 * 3 = 6$. $M = N?$ is $3 = 5$, No.
- Row 5: M increments to 4. $F = F * M = 6 * 4 = 24$. $M = N?$ is $4 = 5$, No.
- Row 6: M increments to 5. $F = F * M = 24 * 5 = 120$. $M = N?$ is $5 = 5$, Yes. Output $F = 120$.

KEY OBSERVATION

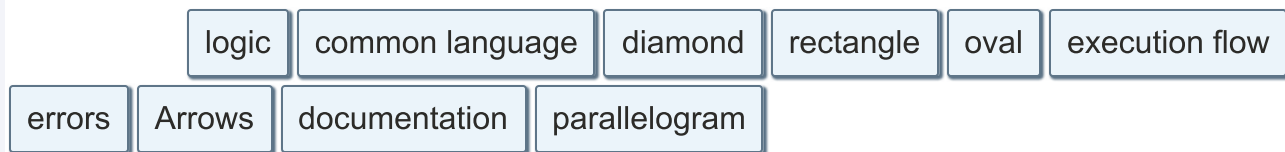
Each pass through the loop produces exactly one row in the trace table. A loop that runs 5 times produces 5 rows of loop output. If you are asked how many times the loop executes, count the rows where the Decision column contains No, then add 1 for the final Yes.

B 1.1.4 Quiz

Test your knowledge with this quiz

1

Fill in the blanks with the appropriate flowchart symbol or concept:



The [oval] symbol represents the start or end of an algorithm.

A [diamond] symbol indicates a decision point where the flow of execution can branch.

The [rectangle] symbol is used to represent a process, such as a calculation or an assignment.

[Arrows] show the direction of flow in a flowchart.

Tracing a flowchart involves following the [execution flow] step-by-step.

The [parallelogram] symbol represents input or output operations.

Flowcharts help visualise the [redacted] **[logic]** of an algorithm.

By tracing a flowchart, you can identify potential [redacted] **[errors]** in your algorithm.

Flowcharts serve as a [redacted] **[common language]** for programmers and non-programmers to discuss algorithms.

Flowcharts are valuable [redacted] **[documentation]** for your code.

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